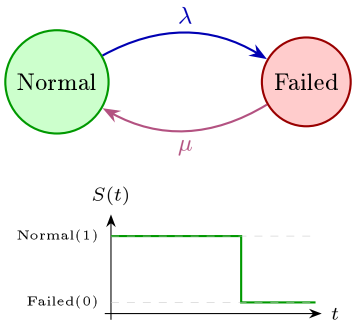
Q8 — Dependability Modeling via Markov Chains

MM10 · Hans-Peter Schwefel

For Analysing Dependability, we look at availability and reliability

1 Availability & Reliability (simple Binary Case)



μ present ⇒ **availability** $A(t)$.
Remove μ ⇒ **reliability** $R(t)$: failed absorbing (Q_{red}).

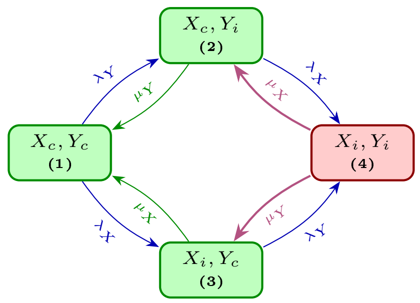
Availability — readiness for correct service:
$A(t) :=$ Pr(sys in normal state at time t)

Reliability — continuity of correct service:
$R(t) :=$ Pr(sys in normal state throughout $[0, t]$)

Assumes exp. distributed time-to-failure & time-to-repair ⇒ Markov property (memoryless).

Key difference: $A(t)$ uses full Q (repair μ present). $R(t)$ uses Q_{red}: failed states absorbing (remove μ transitions). Only model repair for A, not R.

2 2-Component Product State Space Reducant



Transistions:

- Blue λ: failure.
- Green μ: repair (in A and R).
- Magenta μ: **remove for** Q_{red} (state 4 absorbing ⇒ reliability).

States: c=correct, i=incorrect service.
Correct = **{1,2,3}** (X_c or Y_c still serving).
Failed = **{4}** (both X_i and Y_i).

Generator matrix Q (λ=fail rate, μ=repair rate):

$$Q = \begin{pmatrix} -(\lambda_X+\lambda_Y) & \lambda_Y & \lambda_X & 0 \\ \mu_Y & -(\lambda_X+\mu_Y) & 0 & \lambda_X \\ \mu_X & 0 & -(\lambda_Y+\mu_X) & \lambda_Y \\ 0 & \mu_X & \mu_Y & -(\mu_X+\mu_Y) \end{pmatrix}$$

Diagonal $= -\sum_{\text{outgoing rates}}$. Row sums $= 0$.

Transient probabilities:

$$\underline{P}(t) = \underline{P}(0)\,\mathrm{expm}(Qt)$$

Matlab: `P0 * expm(Q*t)`. Start: $\underline{P}(0) = [1, 0, 0, 0]$.

Availability (redundant): $A(t) = P_1(t) + P_2(t) + P_3(t)$

Reliability: same but use Q_{red}: set μ_X, μ_Y in row 4 $= 0$ (state 4 absorbing).

Steady-state: solve $\underline{\pi}Q = \underline{0}$, $\sum \pi_i = 1$.

Note
If **Non-redundant (serial):** correct={1}; failed={2,3,4}.

3 Model Variants

Correlated faults/repairs:

- X fails ⇒ λ_Y **increases** in states 2/3
- Both fail (X_i, Y_i) ⇒ μ_X, μ_Y **degrade** (shared technician, slower repair)

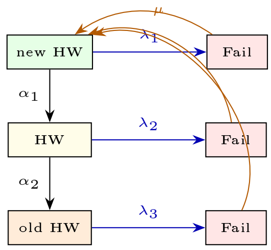
⇒ Modify Q entries differently per state.

Redundancy variants (only *active* comp. fails):

- **Active standby:** both on ⇒ both have λ
- **On demand (instant):** standby λ=0; switch instant
- **On demand (exp. D):** add intermediate switching state

Multiple fault types (HW + SW): rates add: λ=λ_{hw}+λ_{sw}. Repair destination/rate depends on fault type.

4 Non-Constant Failure Rates: Bathtub Model



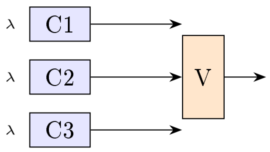
Bathtub hazard rate: $\lambda_1 > \lambda_2 < \lambda_3$

- λ_1 high: early life (production errors)
- λ_2 low: useful life
- λ_3 high: wear-out / aging

Model: 3 HW-quality states each with matching Fail state = 6-state CTMC. Aging: α_1, α_2 transitions (working states degrade over time).

Repair depends on model:
New HW (orange): technician replaces ⇒ μ returns to *new HW* state (same λ as brand new).
Fix only: returns to same HW phase (may also include μ for HW replacement). μ rate and destination depend on what is modelled.

5 TMR: Triple Module Redundancy



"2-out-of-3" with perfect voter (system works if ≥ 2 of 3 OK):

$$R(t) = 3R_c(t)^2 - 2R_c(t)^3, \quad R_c(t) = e^{-\lambda t}$$

$$\mathrm{MTTF}_c = \frac{1}{\lambda} \quad \text{vs.} \quad \mathrm{MTTF}_{\mathrm{TMR}} = \frac{5}{6\lambda}$$

TMR: lower MTTF than single component, but **higher** $R(t)$ **short-term** ⇒ useful for short mission times.

Warning: MTTF alone does not fully characterise dependability — it says nothing about the distribution shape (single-component MTTF > TMR MTTF, yet TMR is more reliable early on).